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Table of Contents

Section	Title	Page
1.0	Introduction	2
2.0	System Rationale	3
2.1	Problem Description	3
2.2	Background	3
2.3	Community Impact	4
2.4	Proposed Solution	4
3.0	Computational Thinking Process	5
3.4	Algorithm Design	5
3.5	SDG Alignment	5
3.6	Open-source Justification	6
4.3	Digital public goal	6
5.0	System Pseudocode	7
6.0	Flowchart	7
6.1	Functional Modules	8
6.2	Expected Output	
7.0	Conclusion	8J
8.0	References	9

1.0 Introduction.

Solid waste management presents one of the most critical urban infrastructural challenges across rapidly expanding cities in developing nations. In Freetown, the capital of Sierra Leone, rapid urban migration and high population density have led to an exponential increase in daily municipal solid waste generation. Unfortunately, municipal administration infrastructure and logistical workflows have struggled to scale proportionally with this mounting waste output.

This report presents the conceptual and logical software architecture for the Kissy Community Waste Collection and Management System (KCWMS). Designed specifically for the high-density Kissy community in Eastern Freetown, KCWMS leverages fundamental principles of computational thinking to streamline waste collection asset tracking, equipment failure reporting, dynamic disposal rerouting, and organic waste diversion. By structuring these real-world operational challenges into clear algorithmic processes, this system establishes a definitive logical blueprint for future software implementations aimed at transforming municipal sanitation management in Sierra Leone.

2. Rationale

2.1 Problem Description

The Kissy community, located in the eastern zone of Freetown, represents an area severely affected by unmanaged municipal waste accumulation. The core problem lies in the structural breakdown of traditional waste collection logistics. Currently, waste collection services operate on manual, uncoordinated schedules without real-time tracking of essential physical assets, such as compactor trucks, skip loaders, and motorized tricycles.

When collection vehicles suffer mechanical failure or when collection sites reach operational bottlenecks, there is no centralized logical mechanism to communicate these breakdowns to management authorities or affected residents. Consequently, waste remains uncollected in public areas for extended periods, creating severe environmental hazards, clogging urban drainage networks, and exposing dense urban populations to severe public health risks. Furthermore, current operational practices treat all municipal refuse as a homogenous mass, failing to separate recoverable organic waste from general refuse, which drastically shortens the functional lifespan of municipal landfill infrastructure.

2.2 Background

The historical and operational background of waste disposal in Eastern Freetown centre around the Kissy Bomeh Dumpsite. Established decades ago, as an unlined open landfill, Kissy Bomeh has long exceeded its planned holding capacity. According to municipal assessments by the Freetown City Council (2024), the site receives hundreds of tons of unsorted municipal waste daily from residential, commercial, and market zones across the eastern corridor.

Major Systemic Infrastructure Gaps Identified:

- Equipment Bottlenecks & Single-Point Failures: The municipal fleet relies heavily on a limited number of heavy mechanical loaders and compactors. When a primary vehicle breaks down, the collection chain in designated wards collapses entirely due to the absence of an automated backup allocation protocol.
- Lack of Real-Time Alternative Rerouting: When access routes to Kissy Bomeh are blocked or when the facility experiences operational halts, collection crews halt operations or dump waste in non-designated transit spots rather than rerouting to secondary sanitary landfills (Sierra Leone Environmental Protection Agency, 2023).
- Unused Organic Resource Potential: Approximately 60% to 70% of municipal waste in Freetown consists of biodegradable organic material originating from household kitchens and vibrant market hubs like Low-Cost Market and Kissy Market. The lack of a systematic sorting workflow leads to rapid waste volume expansion and high methane generation.
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2.3 Community Impact

Developing and deploying an algorithmic management system like KCWMS yields multi-layered, positive socio-economic and environmental impacts across several key groups within the Kissy community.

2.3.1 **Primary Beneficiaries & Impact Analysis**

- Kissy Dockyard & Culvert Residents: Residential areas adjacent to major drainages directly benefit from reduced refuse overflow, eliminated foul Odors, and a significantly lower incidence of vector-borne illnesses (such as malaria, cholera, and typhoid) caused by stagnant water and uncollected garbage heaps.
- Market Traders (Low-Cost, Kissy Market & Trade Centre): Commercial vendors suffer reduced foot traffic when market environments become unsanitary. Timely, verified waste removal maintains clean market premises, protecting food products from contamination and safeguarding informal micro-enterprises.
- Children and Vulnerable Groups: Open dumping exposes playing children to toxic landfill leachate and sharp hazards. Systematized waste collection eliminates illegal neighbourhood dump mounds, creating safer public spaces.
- Freetown City Council (FCC) & Municipal Sanitation Workers: KCWMS equips operational teams with structured digital reporting workflows. Instead of manually searching for available equipment or reacting to public complaints after severe pileups, operators can instantly log machinery failures, automatically alert community stakeholders, and route trucks according to capacity.

2.3.2 Long-Term Environmental & Financial Sustainability

By integrating a dedicated *Organic Waste Diversion and Composting Logic* module, KCWMS fundamentally alters the municipal waste lifecycle. Diverting organic waste into controlled composting channels reduces overall waste volume routed to Kissy Bomeh by up to 50%, significantly extending landfill utility. Furthermore, produced compost provides valuable agricultural inputs for urban farming initiatives in the Western Area, supporting circular economic growth as envisioned in national ICT and environmental assignment guidelines (Limkokwing University of Creative Technology, 2026).

2.4 Proposed Solution

To overcome these challenges, the KCWMS software model introduces five core logical control features:

1. Automated Equipment Verification: Pre-collection status checks to confirm vehicle and loader readiness before dispatch.
2. Instant Failure Logging & Stakeholder Alerting: Immediate status updates dispatched to community leads and FCC supervisors upon machinery breakdown.
3. Dynamic Dumpsite Rerouting Logic: Automatic redirection of collection vehicles to alternative approved landfills when Kissy Bomeh is inaccessible.
4. Organic Waste Separation & Yield Calculation: Embedded mathematical processing to quantify compost potential.
5. Digital Proof-of-Collection Receipting: Automated generation of audit records for completed collection services.

3.0 Computational Thinking Process

Computational thinking involves formulating a problem and expressing its solution in a manner that a computer or a human operational unit can effectively carry out. The design of KCWMS relies on four core computational pillars:

3.1 Decomposition

Decomposition requires breaking down the complex, unstructured problem of city-wide waste management into manageable, discrete functional modules. In KCWMS, the system is decomposed into four operational components: Data Input, Equipment & Facility Validation, Routing & Processing, and Output & Utility Notification.

3.2 Pattern Recognition

By observing recurring collection cycles, we identify consistent operational patterns. Specifically, mixed municipal waste consistently contains high proportions of organic material that follow predictable breakdown rates. Recognizing this pattern allows us to embed a standardized mathematical constant (a 0.5 to 0.7 conversion ratio) to estimate compost output automatically whenever organic waste is logged.

3.3 Abstraction

Abstraction involves focusing strictly on relevant attributes while filtering out non-essential environmental details. For the KCWMS logical model, we retain key variables: userID, communityName, wasteType, wasteWeightKg, equipmentStatus, and kissyBomehStatus, while abstracting away irrelevant details like truck colour, driver identity, and vehicle tire brand.

3.4 Algorithm Design

Algorithm design involves establishing a clear, step-by-step sequence of instructions to produce the correct operational outcome for any valid input state, ensuring predictable execution during both standard operations and hardware failures.

4.0 Algorithm Design & Logic Specification

4.1 Step-by-Step System Algorithm

1. START the execution sequence.
2. Prompt user for credentials (userID, password).
3. Input collection metadata: communityName, dropoffLocation, wasteType, and estimatedQuantity.
4. Evaluate hardware readiness: Check equipmentStatus.
5. IF equipmentStatus is NOT "YES" (Unavailable):

    - Record equipment failure in system audit log.
    - Dispatch notification alerts to FCC operations and community representatives.
    - Set request status to "PENDING - BACKUP REQUIRED". Proceed to step 11.
6. ELSE (Equipment is operational):
    - Evaluate landfill readiness: Check kissyBomehStatus.
    - IF kissyBomehStatus is "OPEN": Set destination to "Kissy Bomeh Dumpsite".
    - ELSE: Set destination to "Alternative Approved Sanitary Landfill".
7. Input actual weighed organic mass: organicWasteWeight.
8. Compute organic compost yield using formula: compostAmount = organicWasteWeight * 0.5.
9. Generate digital Collection Receipt and Compost Report.
10. Display confirmation message: "Collection Process Completed Successfully."
11. END the program sequence.

3.5 Sustainable Development Goals (SDGs) Alignment

The KCWMS project directly supports key United Nations Sustainable Development Goals:

- SDG 11 (Sustainable Cities and Communities): By eliminating uncollected urban garbage piles, preventing stormwater drain blockages, and reducing urban flooding in low-lying Kissy settlements.
- SDG 3 (Good Health and Well-being): By removing disease vectors (mosquitoes, rodents) and toxic runoff, safeguarding public health for market traders and residents.
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3.6 Open-Source Software Justification

Developing KCWMS under an Open-Source software license (such as MIT or GNU GPL) provides critical benefits for Sierra Leone:

- Cost-Effective Scalability: Local municipal councils (FCC, Kenema City Council, Bo City Council) can adopt and customize the software without paying exorbitant foreign license fees.
- Local Tech Capacity Building: Sierra Leonean developers and ICT students can inspect, audit, and enhance the source code, fostering local software engineering skills.
- Community Contributions: Allows non-profit organizations and civic tech groups to contribute localized modules (e.g., SMS failure alerting via local telecom APIs).

3.7 Digital Public Goods (DPGs) Justification

KCWMS fulfills the core requirements of a Digital Public Good by adhering to open standards, promoting public accessibility, and addressing essential societal challenges. By making municipal sanitation data and routing logic openly accessible, KCWMS promotes administrative transparency and enables data-driven urban

planning across developing nations.

4.0 System Pseudocode


```
BEGIN KCWMS_Process
    DECLARE STRING userID, password
    DECLARE STRING communityName, dropoffLocation, wasteType
    DECLARE STRING equipmentStatus, kissyBomehStatus
    DECLARE REAL estimatedQuantity, organicWasteWeight, compostAmount

    INPUT userID, password
    INPUT communityName, dropoffLocation, wasteType, estimatedQuantity
    INPUT equipmentStatus

    IF equipmentStatus == "YES" THEN
        INPUT kissyBomehStatus
        IF kissyBomehStatus == "OPEN" THEN
            PRINT "ROUTE: Direct waste truck to Kissy Bomeh Dumpsite."
        ELSE
            PRINT "ROUTE: Kissy Bomeh unavailable. Redirect to Approved Secondary Landfill."
        ENDIF

        INPUT organicWasteWeight
        compostAmount = organicWasteWeight * 0.5

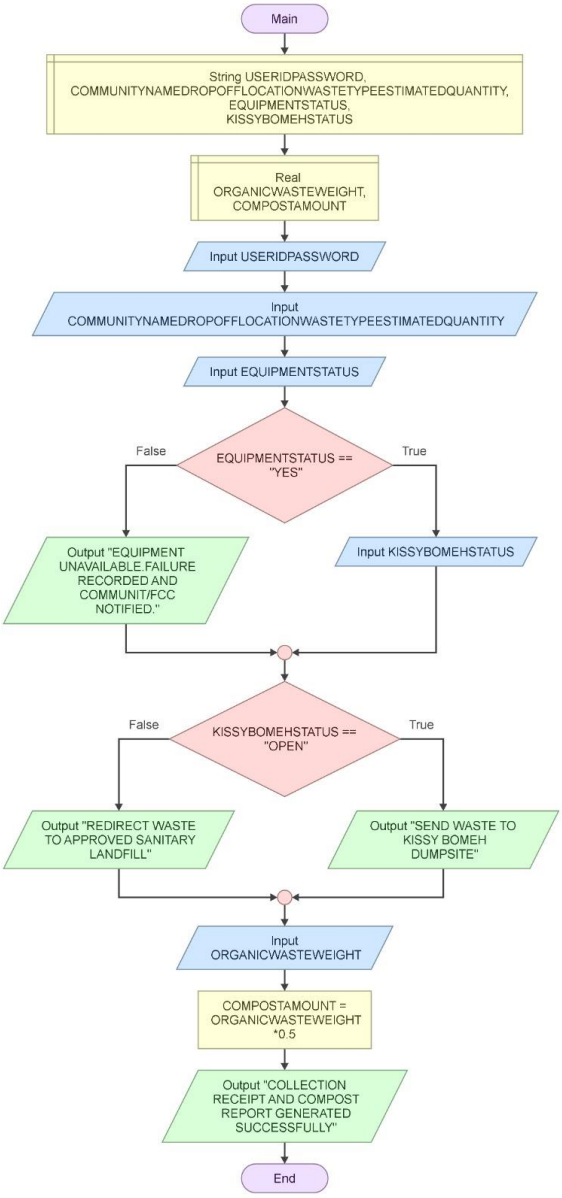
        PRINT "-------------------------------------------------"
        PRINT "OFFICIAL COLLECTION RECEIPT & COMPOST REPORT"
        PRINT "Community Zone: " + communityName
        PRINT "Waste Category: " + wasteType
        PRINT "Processed Organic Mass: " + organicWasteWeight + " kg"
        PRINT "Calculated Compost Yield: " + compostAmount + " kg"
        PRINT "Status: SUCCESSFUL"
```

```
        PRINT "------------------------------------------------"
    ELSE
        PRINT "CRITICAL ALERT: Equipment Unavailable."
        PRINT "ACTION: Equipment failure logged. Automatic alerts sent to Community Reps and FCC Admin."
        PRINT "Status: PENDING BACKUP ALLOCATION"
    ENDIF
END KCWMS_Process
```


4.1 Flowchart Logic & Visual Specification

The system flowchart maps the decision execution paths of the KCWMS logic. It validates user authorization, branches conditionally based on machinery availability (YES/NO), evaluates dumpsite open flags (OPEN/CLOSED), and routes organic calculations accordingly.

KCWMS LOGICAL FLOWCHART ARCHITECTUAL


Main
String USERIDPASSWORD, COMMUNITYNAMEDROPOFFLOCATIONWASTETYPEESTIMATEDQUANTITY, EQUIPMENTSTATUS, KISSYBOMEHSTATUS
Real ORGANICWASTEWEIGHT, COMPOSTAMOUNT
Input USERIDPASSWORD
Input COMMUNITYNAMEDROPOFFLOCATIONWASTETYPEESTIMATEDQUANTITY
Input EQUIPMENTSTATUS
EQUIPMENTSTATUS == "YES"
False
True
Output "EQUIPMENT UNAVAILABLE.FAILURE RECORDED AND COMMUNIT/FCC NOTIFIED."
Input KISSYBOMEHSTATUS
KISSYBOMEHSTATUS == "OPEN"
False
True
Output "REDIRECT WASTE TO APPROVED SANITARY LANDFILL"
Output "SEND WASTE TO KISSY BOMEH DUMPSITE"
Input ORGANICWASTEWEIGHT
COMPOSTAMOUNT = ORGANICWASTEWEIGHT *0.5
Output "COLLECTION RECEIPT AND COMPOST REPORT GENERATED SUCCESSFULLY"
End

5.0 Functional Modules Specification

The KCWMS architectural model is structured into four distinct functional modules to ensure modular software design and ease of future code maintenance:

Module Name	Key Functions	Description & Purpose
1. Input Module	getCommunityRequest() getWasteDetails()	Captures entry data regarding collection locations, estimated waste volumes, operator credentials, and waste categories.
2. Processing Module	checkEquipment() checkKissyStatus() calculateCompost()	Executes logical checks on hardware/facility availability and computes organic decomposition conversion outputs.
3. Routing Module	dumpAtKissy() redirectToNewLandfill()	Determines optimal physical disposal locations based on real-time availability flags.
4. Output & Utility Module	generateReceipt() notifyCommunity() logFailure()	Generates verification documents, triggers administrative notifications, and records equipment failure audit logs.

6.0 Expected System Outputs & Operational Scenarios

6.1 Scenario 1: Standard Operational Flow

Conditions: Equipment available (equipmentStatus = "YES"), Kissy Bomeh open (kissyBomehStatus = "OPEN"), 100 kg organic waste collected at Kissy Dockyard.

- System Routing: Waste dispatched directly to Kissy Bomeh Dumpsite.
- Processing Output: Calculated compost yield = 100 kg * 0.5 = 50 kg.
- Receipt Status: "SUCCESSFUL - DISPATCHED TO KISSY BOMEH".

6.2 Scenario 2: Equipment Failure Event

Conditions: Collection vehicle broken down at Low-Cost Market (equipmentStatus = "NO").

- System Action: Immediate halt of routing sequence; failure logged in operational database.
- Notifications: Broadcast alert dispatched to FCC Maintenance Desk & Low-Cost Market Committee.
- Receipt Status: "PENDING - FAILURE RECORDED - BACKUP REQUESTED".

6.3 Scenario 3: Primary Landfill Closure Event

Conditions: Equipment available (equipmentStatus = "YES"), Kissy Bomeh blocked (kissyBomehStatus = "CLOSED"), 150 kg organic waste at Kissy Culvert.

- System Routing: Dynamic reroute triggered to alternative approved sanitary landfill.
- Processing Output: Calculated compost yield = 150 kg * 0.5 = 75 kg.

- Receipt Status: "SUCCESSFUL - REDIRECTED TO ALTERNATIVE LANDFILL".

7.0 Conclusion

The proposed Kissy Community Waste Collection and Management System (KCWMS) provides a structured, logical solution to the long-standing municipal waste challenges facing the Kissy community in Freetown. By applying fundamental computational thinking principles—decomposition, pattern recognition, abstraction, and algorithm design—the system transforms uncoordinated manual operations into an efficient, predictable digital workflow.

Through real-time equipment monitoring, automated failure reporting, dynamic dumpsite rerouting, and systematic organic waste diversion, KCWMS effectively mitigates environmental health hazards, protects commercial market activities, and provides municipal authorities like the Freetown City Council with actionable administrative visibility. Furthermore, aligning the solution with Open Source principles and Sustainable Development Goals ensures that the system is scalable, transparent, and financially viable for long-term municipal deployment in Sierra Leone.

8.0 References

1. Freetown City Council (FCC). (2024). *Solid Waste Management Report and Byelaws for Municipal Sanitation*. Freetown, Sierra Leone: FCC Press.
2. Limkokwing University of Creative Technology. (2026). *Assignment Brief: Department of Information and Communication Technology, Semester 01 (PROG101: Principles of Programming Logic and Design)*. Freetown Campus, Sierra Leone.
3. Sierra Leone Environmental Protection Agency (SLEPA). (2023). *National Integrated Waste Management Strategy and Environmental Sanitation Framework (2024–2030)*. Freetown, Sierra Leone: EPA-SL Government Publications.